



# PESD12VS1UB

ESD protection diode in SOD523 package

14 September 2026

Product data sheet

## 1. General description

Unidirectional ElectroStatic Discharge (ESD) protection diode in a SOD523 plastic package, designed to protect one transmission or data line from the damage caused by ESD and other transients.

## 2. Features and benefits

- Unidirectional ESD protection of one line
- Low clamping voltage:  $V_{CL} = 35\text{ V}$  at  $I_{PPM} = 5\text{ A}$
- ESD protection up to 30 kV
- IEC 61000-4-5 (surge);  $I_{PPM} = 5\text{ A}$  at  $t_p = 8/20\text{ }\mu\text{s}$

## 3. Application information

- Computers and peripherals
- Communication systems
- Audio and video equipment
- Data lines

## 4. Quick reference data

Table 1. Quick reference data

| Symbol    | Parameter                | Conditions                                                                       | Min | Typ | Max | Unit |
|-----------|--------------------------|----------------------------------------------------------------------------------|-----|-----|-----|------|
| $V_{RWM}$ | reverse standoff voltage | $T_{amb} = 25\text{ }^{\circ}\text{C}$                                           | -   | -   | 12  | V    |
| $C_d$     | diode capacitance        | $f = 1\text{ MHz}$ ; $V_R = 0\text{ V}$ ; $T_{amb} = 25\text{ }^{\circ}\text{C}$ | -   | 38  | 75  | pF   |

5. Pinning information

Table 2. Pinning information

| Pin | Symbol | Description            | Simplified outline                                                                                  | Graphic symbol                                                                                       |
|-----|--------|------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1   | K      | cathode <sup>[1]</sup> | <br>SC-79 (SOD523) | <br><i>sym035</i> |
| 2   | A      | anode                  |                                                                                                     |                                                                                                      |

[1] The marking bar indicates pin 1.

6. Ordering information

Table 3. Ordering information

| Type number | Package |                                                                          |         |
|-------------|---------|--------------------------------------------------------------------------|---------|
|             | Name    | Description                                                              | Version |
| PESD12VS1UB | SC-79   | plastic, surface-mounted package; 2 leads; 1.2 mm x 0.8 mm x 0.6 mm body | SOD523  |

7. Marking

Table 4. Marking codes

| Type number | Marking code |
|-------------|--------------|
| PESD12VS1UB | N3           |

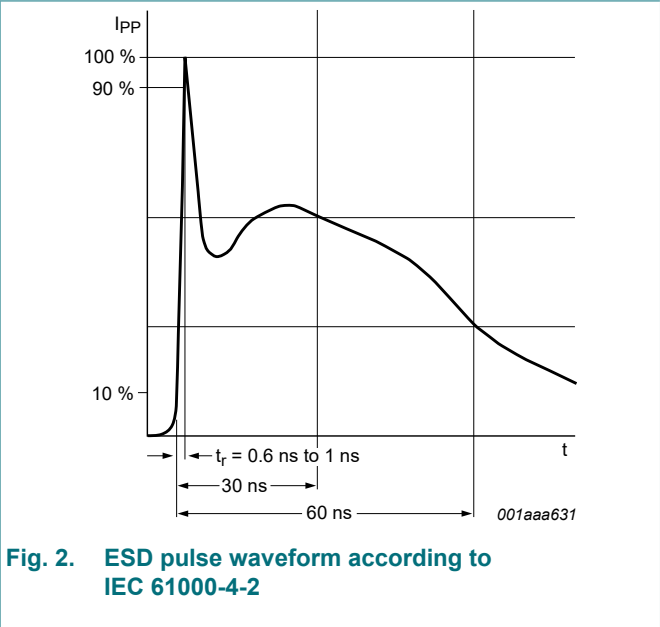
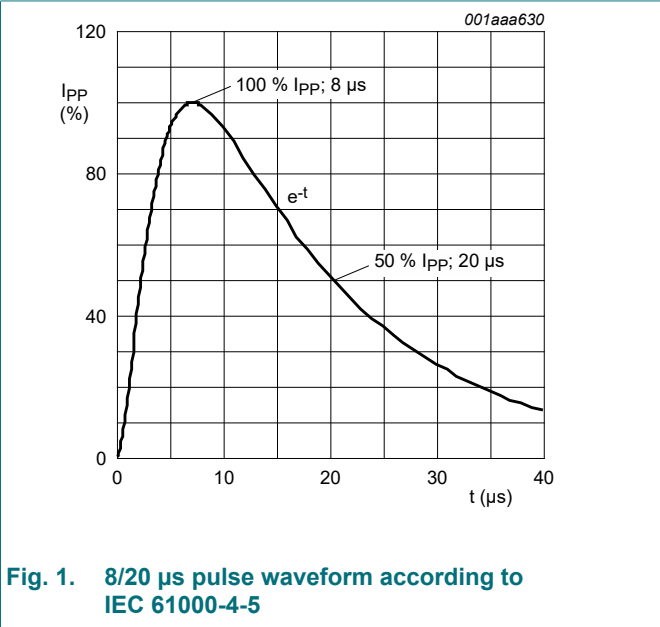
8. Limiting values

Table 5. Limiting values

In accordance with the Absolute Maximum Rating System (IEC 60134).

| Symbol              | Parameter                       | Conditions                          |     | Min | Max | Unit |
|---------------------|---------------------------------|-------------------------------------|-----|-----|-----|------|
| P <sub>PPM</sub>    | rated peak pulse power          | t <sub>p</sub> = 8/20 μs            | [1] | -   | 180 | W    |
| I <sub>PPM</sub>    | rated peak pulse current        | t <sub>p</sub> = 8/20 μs            | [1] | -   | 5   | A    |
| T <sub>j</sub>      | junction temperature            |                                     |     | -   | 150 | °C   |
| T <sub>amb</sub>    | ambient temperature             |                                     |     | -55 | 150 | °C   |
| T <sub>stg</sub>    | storage temperature             |                                     |     | -65 | 150 | °C   |
| ESD maximum ratings |                                 |                                     |     |     |     |      |
| V <sub>ESD</sub>    | electrostatic discharge voltage | IEC 61000-4-2; contact discharge    | [2] | -   | 30  | kV   |
|                     |                                 | MIL-STD-883; human body model (HBM) |     | -   | 10  | kV   |

- [1] Non-repetitive current pulse 8/20 μs exponentially decay waveform.  
[2] Device stressed with ten non-repetitive ESD pulses.

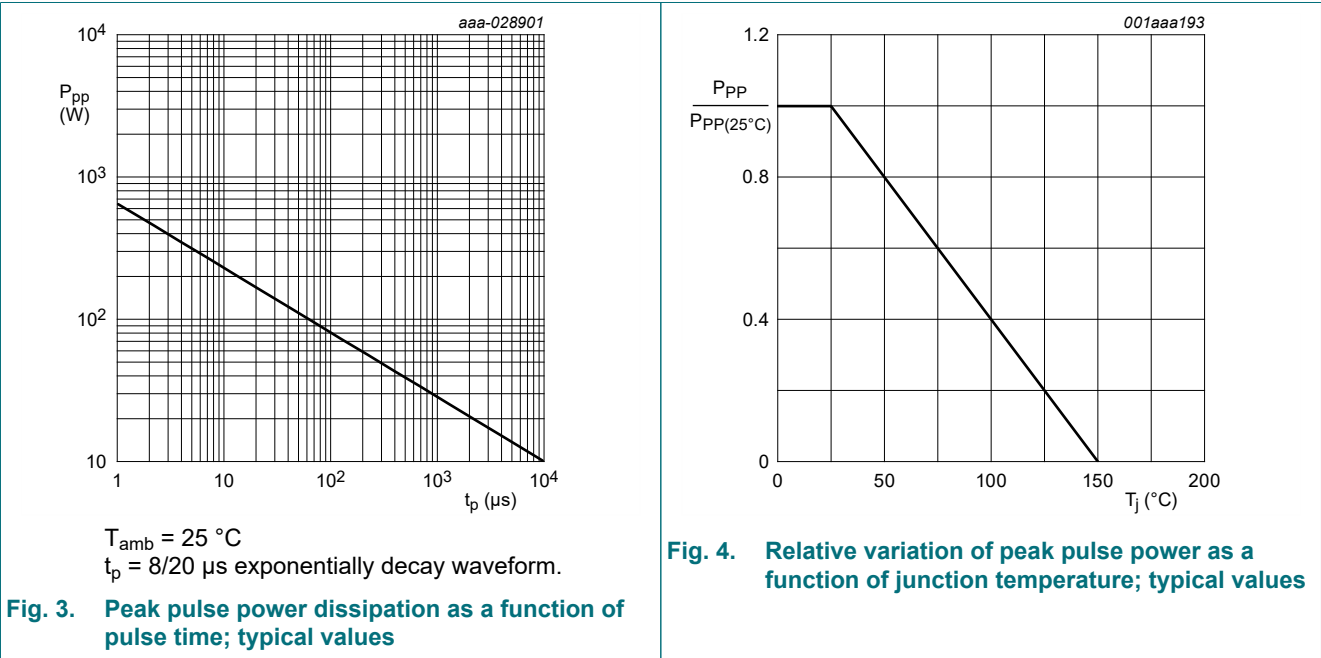


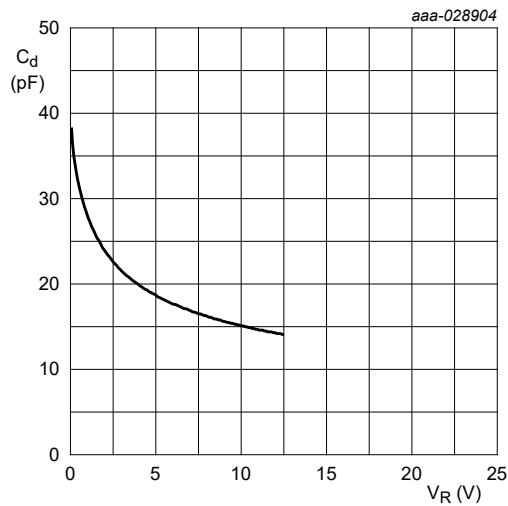
9. Characteristics

Table 6. Characteristics

| Symbol            | Parameter                | Conditions                                                |     | Min  | Typ | Max  | Unit |
|-------------------|--------------------------|-----------------------------------------------------------|-----|------|-----|------|------|
| V <sub>RWM</sub>  | reverse standoff voltage | T <sub>amb</sub> = 25 °C                                  |     | -    | -   | 12   | V    |
| V <sub>BR</sub>   | breakdown voltage        | I <sub>R</sub> = 5 mA; T <sub>amb</sub> = 25 °C           |     | 14.7 | 15  | 15.3 | V    |
| I <sub>RM</sub>   | reverse leakage current  | V <sub>RWM</sub> = 12 V; T <sub>amb</sub> = 25 °C         |     | -    | 1   | 50   | nA   |
| C <sub>d</sub>    | diode capacitance        | f = 1 MHz; V <sub>R</sub> = 0 V; T <sub>amb</sub> = 25 °C |     | -    | 38  | 75   | pF   |
| V <sub>CL</sub>   | clamping voltage         | I <sub>PP</sub> = 1 A; T <sub>amb</sub> = 25 °C           | [1] | -    | -   | 19   | V    |
|                   |                          | I <sub>PPM</sub> = 5 A; T <sub>amb</sub> = 25 °C          | [1] | -    | -   | 35   | V    |
| R <sub>diff</sub> | differential resistance  | I <sub>R</sub> = 1 mA; T <sub>amb</sub> = 25 °C           |     | -    | -   | 200  | Ω    |

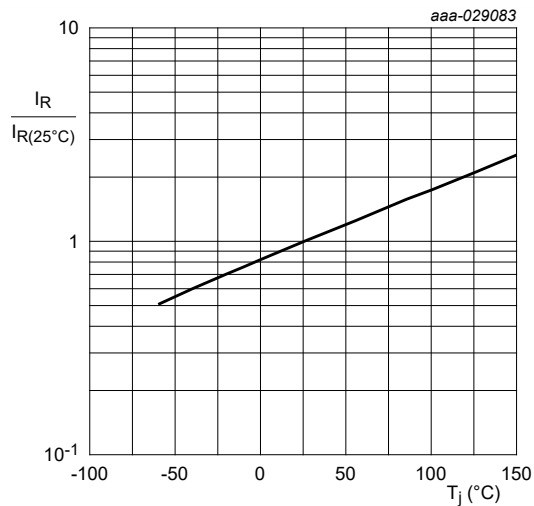
[1] Non-repetitive current pulse 8/20 μs exponentially decay waveform.





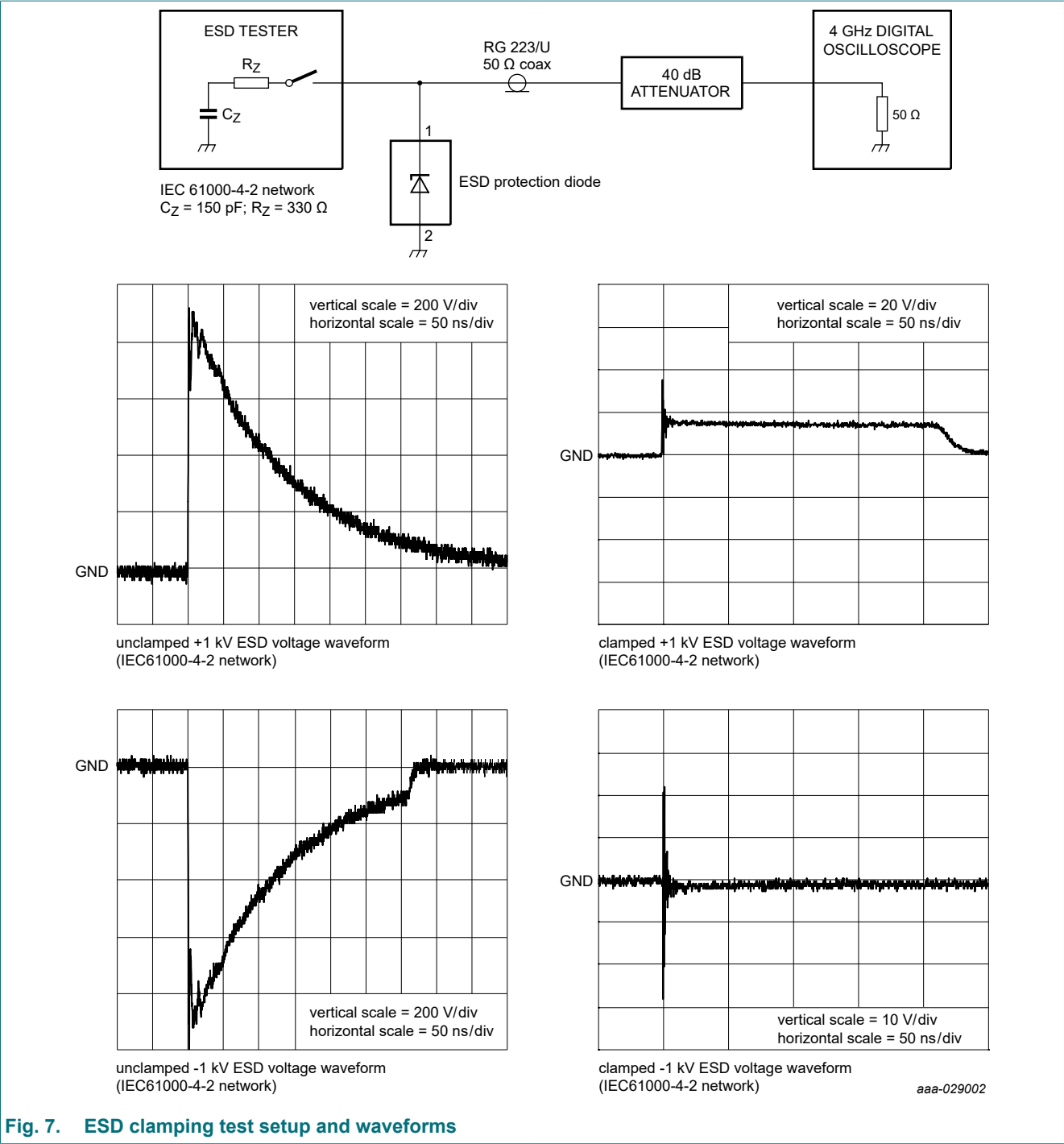
$f = 1\text{ MHz}; T_{\text{amb}} = 25\text{ }^{\circ}\text{C}$

Fig. 5. Diode capacitance as a function of reverse voltage; typical values



$I_R$  is less than 10 nA at 150  $^{\circ}\text{C}$   
 $V_{\text{RWM}} = 12\text{ V}$

Fig. 6. Relative variation of reverse leakage current as a function of junction temperature; typical values



## 10. Application information

The device is designed for protection of one unidirectional data or signal line from surge pulses and ESD damage. The device is suitable on lines where the signal polarities are either positive or negative with respect to ground.

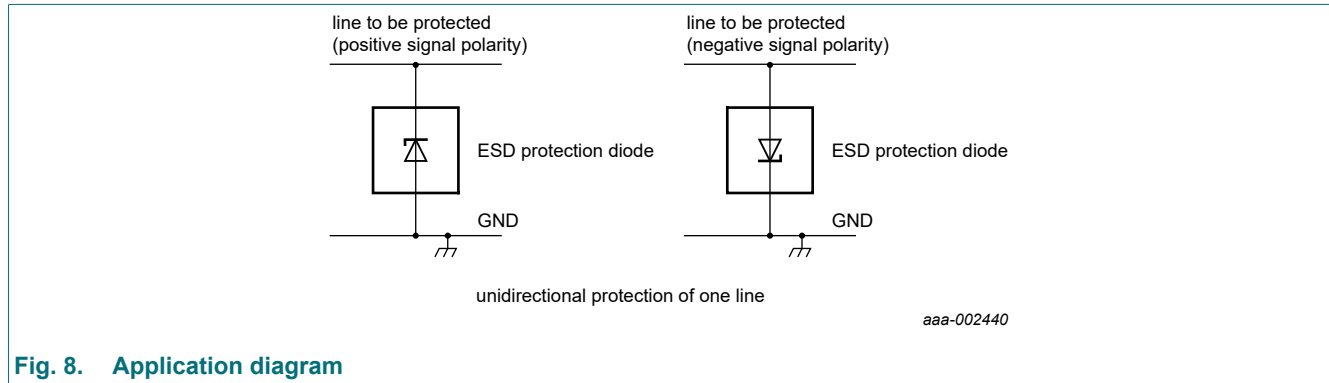


Fig. 8. Application diagram

### Circuit board layout and protection device placement

Circuit board layout is critical for the suppression of ESD, Electrical Fast Transient (EFT) and surge transients. The following guidelines are recommended:

1. Place the device as close to the input terminal or connector as possible.
2. Minimize the path length between the device and the protected line.
3. Keep parallel signal paths to a minimum.
4. Avoid running protected conductors in parallel with unprotected conductors.
5. Minimize all Printed-Circuit Board (PCB) conductive loops including power and ground loops.
6. Minimize the length of the transient return path to ground.
7. Avoid using shared transient return paths to a common ground point.
8. Use ground planes whenever possible. For multilayer PCBs, use ground vias.

11. Package outline

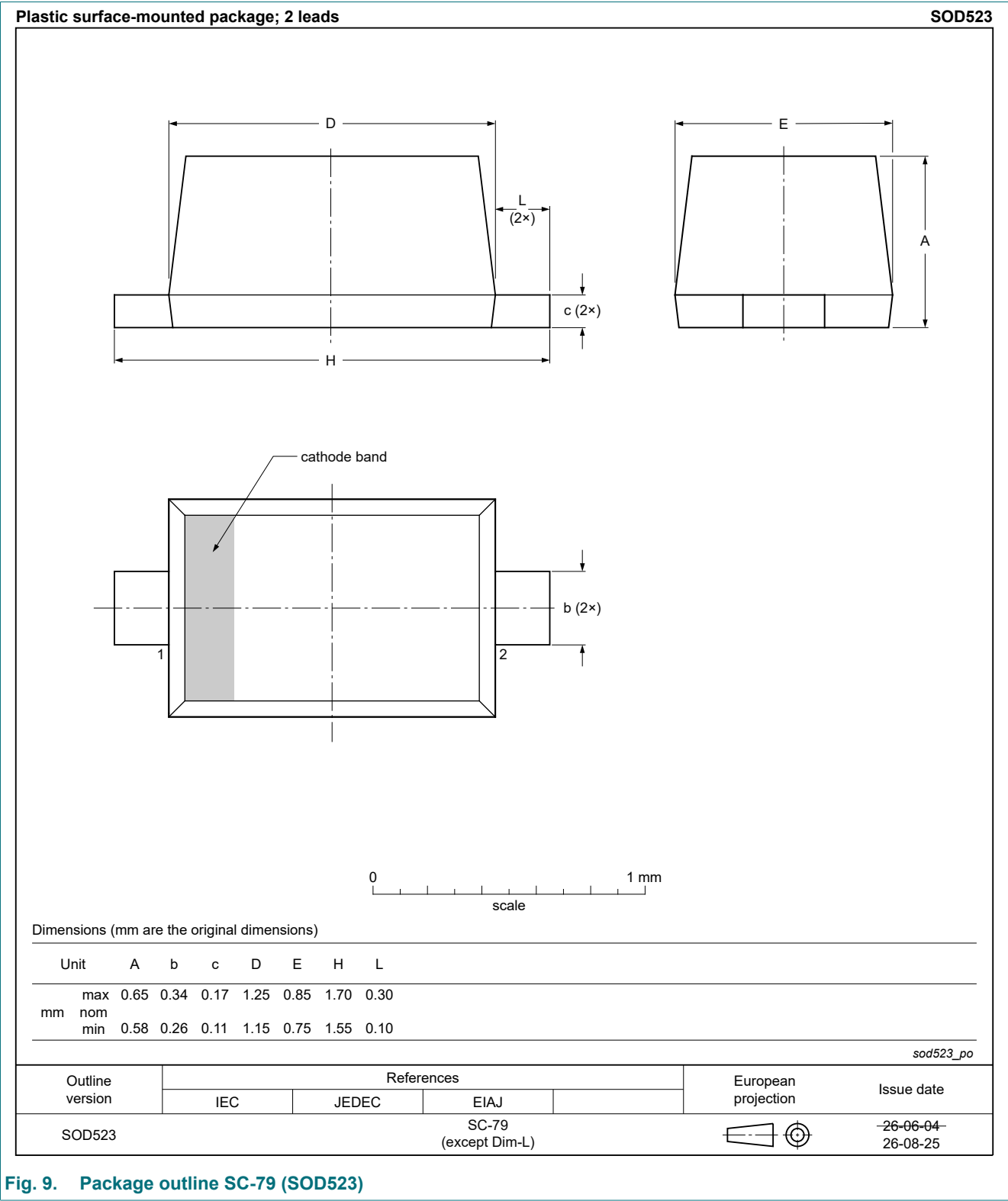
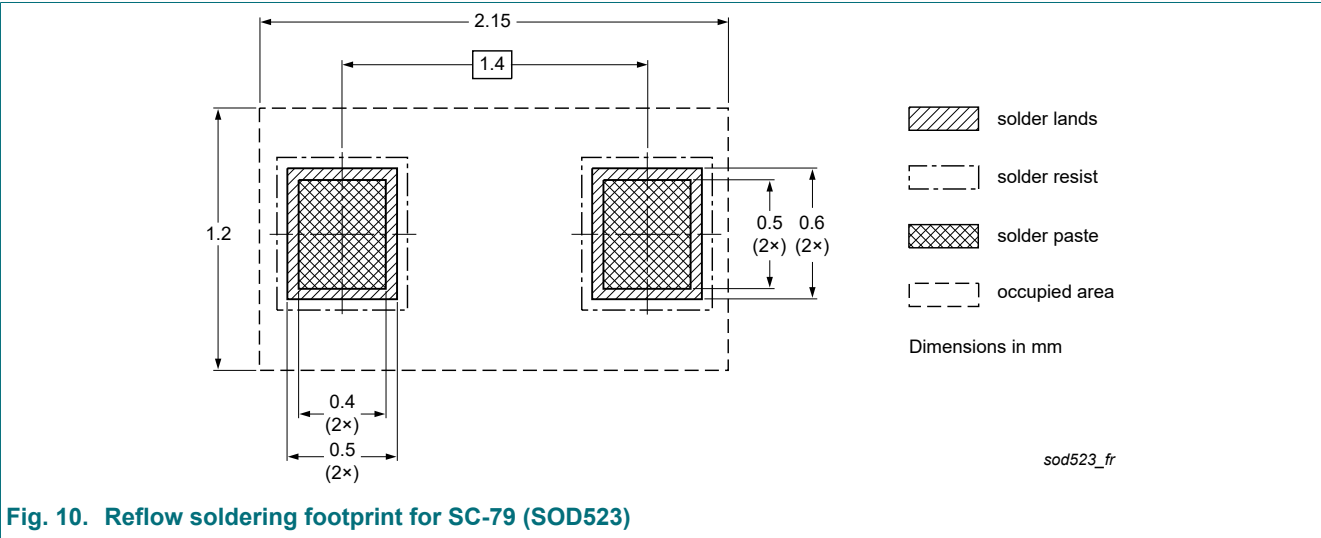


Fig. 9. Package outline SC-79 (SOD523)

12. Soldering



### 13. Revision history

Table 7. Revision history

| Data sheet ID      | Release date                      | Data sheet status  | Change notice | Supersedes         |
|--------------------|-----------------------------------|--------------------|---------------|--------------------|
| PESD12VS1UB v.6    | 20260914                          | Product data sheet | -             | PESD12VS1UB v.5    |
| Modifications:     | • Package outline: Typo corrected |                    |               |                    |
| PESD12VS1UB v.5    | 20260806                          | Product data sheet | -             | PESD12VS1UB v.4    |
| PESD12VS1UB v.4    | 20240502                          | Product data sheet | -             | PESD12VS1UB v.3    |
| PESD12VS1UB v.3    | 20181129                          | Product data sheet | -             | PESDXS1UB_SERIES_2 |
| PESDXS1UB_SERIES_2 | 20090824                          | Product data sheet | -             | PESDXS1UB_SERIES_1 |
| PESDXS1UB_SERIES_1 | 20040614                          | Product data sheet | -             | -                  |

## 14. Legal information

### Data sheet status

| Document status [1][2]         | Product status [3] | Definition                                                                            |
|--------------------------------|--------------------|---------------------------------------------------------------------------------------|
| Objective [short] data sheet   | Development        | This document contains data from the objective specification for product development. |
| Preliminary [short] data sheet | Qualification      | This document contains data from the preliminary specification.                       |
| Product [short] data sheet     | Production         | This document contains the product specification.                                     |

- [1] Please consult the most recently issued document before initiating or completing a design.
- [2] The term 'short data sheet' is explained in section "Definitions".
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